Software Engineering & Design Principal Lab

Experiment 2

Experiment: Identification of Problem, Objectives, Scope and Feasibility Analysis

Student/Team Details

Particular	Details
Experiment No.	2
Project Title	Alumni Management Portal
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Semester & Section	V-D
Date of Submission	21 August 2026

1. Problem Identification

1.1 Background / Context

The proposed project belongs to the Education / Alumni Relations domain. It targets colleges and universities that need a structured way to stay connected with their graduated students (alumni) and to connect those alumni with currently enrolled students for mentorship, job referrals, events and fundraising.

1.2 Existing Problem

Most institutions rely on informal channels — WhatsApp groups, personal Facebook groups, spreadsheets maintained by the alumni cell, and word-of-mouth — to stay in touch with alumni. This makes it difficult to verify who is actually an alumnus, to search alumni by batch, branch, company or location, to advertise job openings or events to the right audience, or to track donations and engagement in one place.

1.3 Existing Method / Current System

Alumni records are currently maintained manually in spreadsheets by the alumni-relations cell. Event invitations and job openings are circulated through email blasts and social-media groups. Donations, where collected, are tracked in separate spreadsheets or ledger books with no online payment integration.

1.4 Limitations of the Existing System

1. Alumni data is scattered across spreadsheets, email threads and social-media groups, with no single verified source of truth.

2. There is no way to verify that a person claiming to be an alumnus is genuine before they access the community.

3. Alumni cannot be searched or filtered by batch, branch, company or location, making it hard to find the right mentor or contact.

4. Job/internship referrals depend on informal messages and are easily missed or lost in group chats.

5. There is no centralized way to track event RSVPs, mentorship requests or donation records, so engagement cannot be measured or reported on.

2. Problem Statement

Who is facing the problem?

- What is the problem?
- What are the limitations of the existing approach?
- Why is a software-based solution required?

Problem Statement

College alumni, current students and the alumni-relations administration face the problem of fragmented, unverifiable and unsearchable alumni information spread across spreadsheets, email threads and informal social-media groups. This makes it difficult to verify alumni identity, discover relevant alumni for mentorship or referrals, publicize events and job openings to the right audience, and track donations or engagement. The existing manual approach does not scale as the alumni base grows and provides no analytics, notifications or accountability. A dedicated, role-based software solution — the Alumni Management Portal — is required to centralize alumni records, automate verification and notifications, and provide dedicated modules for events, jobs, mentorship and donations.

3. Objectives of the Project

S. No. Objective

1. To develop a centralized, role-based web portal that connects alumni, students and college administrators.

2. To automate alumni registration and admin-verified account approval.

3. To provide a searchable, filterable alumni directory by batch, branch, company and location.

4. To enable alumni to post job/internship openings and students to apply directly through the portal.

5. To facilitate a structured mentorship request-and-response workflow between alumni and students.

6. To provide an event management module supporting event creation, discovery and RSVP tracking.

7. To enable secure online donations with automated receipt generation and reporting.

8. To generate real-time in-app and email notifications for key actions across all modules.

4. Scope of the Project

4.1 Target Users / Stakeholders

S. No.	User / Stakeholder	Role
1.	Guest	Unauthenticated visitor who can view public pages and register for an account.
2.	Student	Currently enrolled student who searches the directory, applies to jobs, requests mentorship and RSVPs to events.
3.	Alumni	Verified graduate who posts jobs, creates events, offers mentorship and donates.
4.	Admin	College alumni-relations staff who verify accounts, moderate content and view analytics.

4.2 Functional Scope

1. User registration, role-based login and admin-verified alumni approval.
2. Alumni profile management and searchable alumni directory.
3. Event creation, discovery and RSVP tracking.
4. Job/internship posting and application management.
5. Mentorship request-and-response workflow between alumni and students.
6. Donation and fundraising module with payment-gateway integration.
7. Discussion forum, announcements and a unified notification system.

4.3 Data Scope

S. No.	Data / Information	Description
1.	User & Profile Data	Name, email, role, batch, branch, company, designation, profile photo.
2.	Event Data	Event title, description, date/time, venue, RSVP records.
3.	Job & Application Data	Job postings, applicant resumes, application status.
4.	Mentorship Data	Mentor availability, mentorship requests and their status.
5.	Donation & Transaction Data	Campaign details, donation amount, transaction ID, receipts.

4.4 Out of Scope

1. A native mobile application (the portal is a responsive web application only).
2. AI-based alumni-job or alumni-mentee matching.
3. Multi-institution / multi-tenant support for more than one college.
4. A public REST API for third-party integration (e.g., with the institution's ERP).

5. Technical Feasibility Analysis

5.1 Proposed Technology Stack

Component	Proposed Technology
Frontend	Next.js (React) with Tailwind CSS
Backend	Next.js API Routes (Node.js runtime)
Database	PostgreSQL (Neon serverless Postgres)
Programming Language(s)	TypeScript, JavaScript
Development Tools/IDE	VS Code / Antigravity, Git & GitHub, Postman
Other Technologies	Prisma ORM, NextAuth.js, Cloudinary, Razorpay/Stripe payment gateway, Vercel hosting

5.2 Hardware Requirements

Hardware	Requirement
Processor	Intel Core i5 (10th Gen) or equivalent — development machine only; end users need no special hardware
RAM	8 GB minimum, 16 GB recommended for development
Storage	10 GB free SSD space for the codebase, dependencies and local tooling
Other Hardware	A client device with a modern web browser and a stable internet connection

5.3 Software Requirements

Software	Version / Requirement
Operating System	Windows 10/11, macOS or Linux (development); any OS with a modern browser (client)
IDE / Editor	Visual Studio Code / Antigravity
Database	PostgreSQL 15+ (managed via Neon)
Framework / Libraries	Next.js 14, Prisma ORM, Tailwind CSS, NextAuth.js, Zod
Other Software	Node.js 18+, Git, npm, Postman

5.4 Technical Skills Required

React.js / Next.js and component-based UI development

- Node.js and RESTful API design
- SQL and ORM-based database design (Prisma)
- Git/GitHub version control and basic deployment (Vercel)

Technical Feasibility Assessment

All required technologies are open-source or available on free developer tiers (Next.js, PostgreSQL via Neon, Prisma, Cloudinary, Vercel), so there is no licensing cost barrier. The developer already has hands-on experience with this exact stack from prior projects — React/Next.js certifications, MERN-stack projects, and an AWS Certified Cloud Practitioner certification — and has previously built and deployed projects using Next.js, ORM-based database access and token-based authentication. The development machine and internet connectivity available are sufficient to build, test and deploy the system within the semester timeline.

Conclusion

Technically Feasible

Justification:

The required technology stack is well-documented, free or low-cost to use, and already familiar to the developer, so the project can realistically be built and deployed with the available resources and timeline.

6. Operational Feasibility Analysis

Operational feasibility determines whether the proposed system can be effectively used and integrated into the intended environment.

6.1 User Acceptance

Students and alumni are already comfortable using web and social platforms such as LinkedIn, Facebook groups and job portals, so a similarly structured, role-based directory-and-job-board interface is expected to be readily accepted. The Admin (alumni-relations staff) is expected to welcome the system since it replaces manual spreadsheet upkeep with an automated, auditable workflow.

6.2 Ease of Use

The portal uses a clean, role-based dashboard so each user sees only the actions relevant to them — for example, Students see “Apply” and “Request Mentorship”, while Alumni see “Post a Job” and “Create Event”. Search, filters and forms follow familiar web-application patterns, keeping the learning curve minimal.

6.3 Existing Workflow

The system does not replace the alumni-relations cell's role — it digitizes and centralizes the verification, communication and record-keeping steps they already perform manually, fitting into the existing process while removing the spreadsheet- and email-based bottlenecks.

6.4 Training Requirements

Students and Alumni require no formal training beyond a short onboarding tour, as the interface follows common web-application conventions. Admin users will be given a brief walkthrough / user guide covering account verification, event/announcement posting and report export.

6.5 Expected Operational Benefits

1. A single, verified, searchable alumni database in place of scattered spreadsheets.

2. Higher event participation through in-app and email RSVP notifications.
3. Faster job and internship referrals directly between alumni and students.
4. Transparent, reportable donation tracking with automated receipts.

Operational Feasibility Assessment

Operationally Feasible

Justification:

The system fits naturally into the alumni cell's existing responsibilities, requires minimal training, and mirrors interaction patterns users are already familiar with from other web platforms, so it is expected to be adopted smoothly.

7. Overall Feasibility Summary

Feasibility Area	Status	Justification
Technical Feasibility	Feasible	Familiar, free/open-source technology stack matched to the developer's existing skills and the project timeline.
Operational Feasibility	Feasible	Low training overhead, digitizes an existing manual workflow, and mirrors familiar web-application interaction patterns.

Overall Conclusion

Based on the technical and operational feasibility analysis, the proposed project is **feasible** because the required technologies are freely available and already familiar to the developer, and the system fits naturally into the alumni-relations cell's existing workflow with minimal training or adoption risk.

8. Expected Outcome

The completed Alumni Management Portal is expected to provide a single, secure, role-based platform where alumni are verified and searchable, events and job/internship opportunities reach the right audience, mentorship connections can be requested and tracked, and donations can be collected and reported on — replacing the institution's current fragmented, manual approach to alumni engagement.

9. Future Enhancements *(Optional)*

1. Real-time notifications via WebSocket/Server-Sent Events instead of load-time polling.
2. AI-assisted alumni–job or alumni–mentee matching.
3. A native mobile application and a public API for integration with the institution's ERP.